

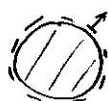
NS3

A spherical balloon of variable radius $R(t)$ is immersed in an infinite liquid medium of density ρ and viscosity μ . Far from the balloon, the liquid is at rest with pressure $p = p_\infty$. Determine the radial velocity $v_r(r, t)$ and pressure distribution $p(r, t)$ for the buoyancy-free spherically-symmetrical motion induced in the surrounding liquid. Obtain also the power required to inflate the balloon.

CONT: $\frac{1}{r^2} \frac{d}{dt} (r^2 v_r) = 0 \rightarrow r^2 v_r = C(t) \xrightarrow{r=R} v_r = \frac{dR}{dt} = \dot{R} \Rightarrow \boxed{v_r = \frac{R^2 \dot{R}}{r^2}}$

r-MOM: $\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} \right) = - \frac{\partial p}{\partial r} \Rightarrow \frac{\partial p}{\partial r} = - \rho \left(\frac{2 R \ddot{R} + R^2 \dot{R}}{r^2} - \frac{2 R^4 \dot{R}^2}{r^5} \right) \Rightarrow \frac{p - p_\infty}{\rho} = \frac{2 R \ddot{R} + R^2 \dot{R}}{r} - \frac{1}{2} \frac{R^4 \dot{R}^2}{r^4}$

$\dot{W}_{\text{INFLATE}} = \dot{W}_{B \rightarrow l} = - \dot{W}_{l \rightarrow B}$



$\dot{W}_{l \rightarrow B} = - \int_{\Sigma_B} p \vec{v} \cdot \vec{n} d\sigma + \int_{\Sigma_B} \vec{v} \cdot \vec{\tau} \cdot \vec{n} d\sigma = - p(R) \dot{R} 4\pi R^2 - 4\mu \dot{R} 4\pi R^2$
 $\vec{v} \cdot \vec{\tau} \cdot \vec{n} = v_r \tau_{rr} = -4\mu \frac{\dot{R}}{R}$
 $\int 2\mu \frac{\partial v_r}{\partial r} = -4\mu \frac{R^2 \dot{R}}{r^3}$

$\dot{W}_{\text{INFLATE}} = \dot{W}_{B \rightarrow l} = - \dot{W}_{l \rightarrow B} = p_\infty 4\pi R^2 \dot{R} + 4\pi \rho R^2 \dot{R} \left(\frac{3}{2} \dot{R}^2 + R \ddot{R} \right) + 16\pi \mu R \dot{R}^2$

ADDITIONAL QUESTION: WHERE DOES THE WORK GO? $\Phi_r = \mu \left\{ 2 \left[\left(\frac{\partial v_r}{\partial r} \right)^2 + \frac{v_r^2}{r^2} \right] \right\} = 12\mu \frac{R^4 \dot{R}^2}{r^6}$
 CONSIDER THE FOLLOWING CONTROL VOLUME

$\frac{d}{dt} \int_{V_c} \rho \frac{v^2}{2} dV + \int_{\Sigma_R} \rho \frac{v^2}{2} (\vec{v} \cdot \vec{e}_r) \cdot \vec{n} d\sigma = - \int_{\Sigma_R} p \vec{v} \cdot \vec{n} d\sigma + \int_{\Sigma_R} \vec{v} \cdot \vec{\tau} \cdot \vec{n} d\sigma + \int_{\Sigma_R} \rho \vec{v} \cdot \vec{\tau} \cdot \vec{n} d\sigma - \left[- \int_{V_c} p \vec{v} \cdot \vec{n} d\sigma + \int_{V_c} \Phi_r dV \right]$
 $\downarrow$
 $4\pi \rho R^2 \dot{R} \left(\frac{3}{2} \dot{R}^2 + R \ddot{R} \right) = \dot{W}_{B \rightarrow l} - 4\pi R^2 \dot{R} p_\infty - 16\pi \mu R \dot{R}^2$

$\dot{W}_{B \rightarrow l} = \underbrace{\frac{d}{dt} \int_{V_c} \rho \frac{v^2}{2} dV}_{\text{INCREMENT OF MECHANICAL ENERGY}} + \underbrace{\int_{\Sigma_\infty} p \vec{v} \cdot \vec{n} d\sigma}_{\text{PRESSURE WORK NEEDED TO SET THE AMBIENT GAS IN MOTION}} + \underbrace{\int_{V_c} \Phi_r dV}_{\text{MECHANICAL ENERGY CONVERTED INTO INTERNAL ENERGY DUE TO VISCOUS DISSIPATION}}$